

Winfomi FieldTrack

Tech Stack & Integrations

Tech Stack

Mobile App (Field Rep)

Layer	Technology	Version
Framework	React Native	0.81.5
Runtime/Build	Expo (Managed)	~54.0.37
OTA Updates	Expo Updates (expo-updates)	~29.0.20
Build & Distribution	EAS (Expo Application Services) CLI	22.6.0
Navigation	React Navigation (native-stack, bottom-tabs)	^7.x

Key Expo & Native Modules

Purpose	Library
GPS location / geofencing	expo-location (~19.0.8)
Secure token storage	expo-secure-store (~15.0.8)
Push notifications	expo-notifications (~0.32.17)
Background location	expo-task-manager (~14.0.9)
Splash / system UI	expo-splash-screen, expo-system-ui
Network state (offline queue)	@react-native-community/netinfo
Local key-value storage	@react-native-async-storage/async-storage
Calendar / date picker	react-native-calendars, @react-native-community/datetimepicker
HTTP client	axios
Icons	lucide-react-native

Backend

Layer	Technology
Language / Framework	Python + FastAPI
Hosting	Render (Mumbai-adjacent region, free web service plan)
Database	PostgreSQL, via asyncpg connection pool (raw SQL, no ORM)
Auth	JWT (PyJWT), bcrypt password hashing
Scheduled jobs	APScheduler (auto-cutoff, absence-check, idempotency cleanup, employee-state cache sweep)
Rate limiting	slowapi (per-IP, trust-proxy hop-aware)
Security headers	Custom middleware reproducing helmet's default header set
Geocoding / routing	Google Maps Platform: Geocoding, Places, Routes (Compute Routes) APIs
Server	uvicorn (ASGI)

The backend is a REST API server that handles: JWT-based login/refresh for reps and managers; day check-in/check-out and dealer check-in/check-out with GPS geofence validation; dealer visit assignments and navigation (Google Routes-backed "tap to navigate"); manager dashboard data, reports (CSV export), and notifications; offline-sync request replay with Idempotency-Key deduplication; and two self-scheduling background sweeps (auto-cutoff for reps who never log out, absence-check for reps who never log in).

Web Dashboard (Manager)

Layer	Technology
Framework	Next.js (React)
Hosting	Vercel, auto-deployed from the main branch
Maps	Google Maps JavaScript API (dealer/rep location map)
HTTP client	axios
Icons	lucide-react
Testing	vitest + @testing-library/react

Database / External Data

Service	Purpose	Connection
PostgreSQL	Employees, dealers, attendance, visits, assignments, navigations, notifications, notes, reminders, sync-failure log	DB_* env vars -> asyncpg connection pool
Google Maps Platform	Reverse geocoding, place search, driving distance/duration (Routes API)	Server-side API key (GOOGLE_MAPS_API_KEY)
Google Maps JavaScript API	Dealer/rep location map on the web dashboard	Client-side, referrer-restricted API key
Local Device Storage (mobile)	Offline-first attendance/visit queue, cached today-state	expo-secure-store, AsyncStorage

Cloud Services

Service	Provider	Purpose
App Hosting (Backend)	Render	Hosts the FastAPI REST API
Database Hosting	Render (managed PostgreSQL)	Stores all application data
Web Hosting	Vercel	Hosts the Next.js manager dashboard
Maps & Geocoding	Google Maps Platform	Geocoding, Places, Routes APIs
OTA Updates	Expo EAS (expo.dev)	Push JS bundle updates to installed apps without an app-store review
Mobile Builds	Expo EAS Build	Produces installable Android APKs / app-store builds

How Everything Connects

1. Authentication - JWT

The backend issues its own JWTs on login (no third-party identity provider) - a rep or manager signs in with a username/password, bcrypt-verified against the employees table.

- The app calls POST /api/auth/login with a username and password.
- The backend verifies the password hash and returns a short-lived access token and a longer-lived refresh token.
- The access token is persisted via expo-secure-store on mobile (localStorage-backed session on web) and sent as a Bearer token on every subsequent API call.
- When the access token expires, the app calls POST /api/auth/refresh with the refresh token to get a new pair without forcing the user to log in again.
- Every protected route validates the JWT's signature, expiry, and role claim (rep vs manager) before running any business logic.

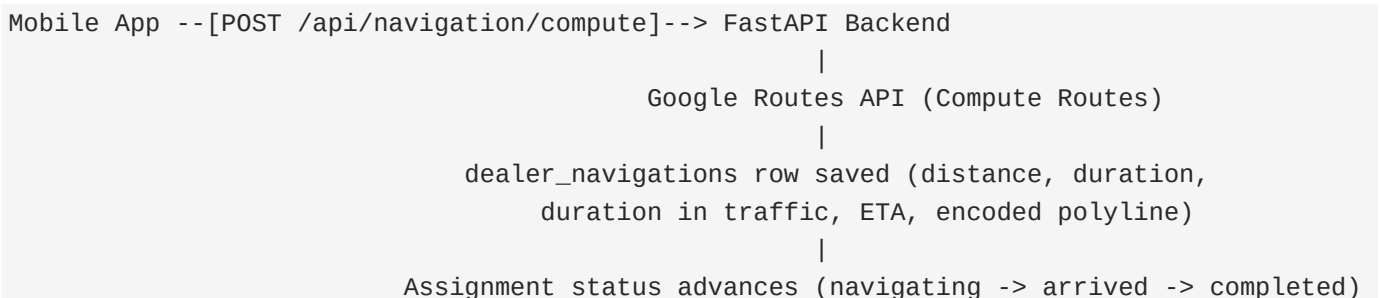
2. Attendance - Day & Dealer Check-in/Check-out

The core of the app: a rep's working day and each dealer visit within it are tracked as separate, GPS-validated events.

- Day check-in - rep taps "Start Day"; the app captures GPS coordinates via expo-location and sends them to POST /api/attendance/checkin.
- Dealer check-in/check-out - for an assigned dealer, the app captures GPS and calls POST /api/visits/login and POST /api/visits/logout; the backend haversine-checks the coordinates against the dealer's registered location and radius before accepting the visit.
- Day check-out - POST /api/attendance/checkout ends the day; an APScheduler sweep also auto-cuts-off any day left open past a configured cutoff hour.
- Offline handling - if a check-in/check-out request fails due to no network, it's queued locally with an Idempotency-Key and replayed automatically on reconnect, so the same action is never double-recorded server-side.

3. Dealer Navigation - "Tap Navigate"

When a rep taps Navigate on an assigned dealer, the backend calls Google's Routes API (Compute Routes) for a real driving distance/duration/ETA and polyline - never a Route Optimization/Fleet Routing call, always a single origin-to-destination pair matching the fixed assignment sequence.



- PATCH /api/navigation/{id}/status updates the navigation's lifecycle as the rep arrives, completes, or cancels the trip.
- GET /api/navigation/summary/today powers the rep's Daily Travel Summary (distance travelled, distance remaining, driving time).

4. Manager Dashboard & Map

- GET /api/dashboard/today returns each rep's live status (logged in / not logged in / day ended), visit counts, and distance travelled - the web dashboard polls/refreshes this on load.
- GET /api/dashboard/map returns rep and dealer coordinates for the manager map view, rendered with the Google Maps JavaScript API, with dealer/rep search.
- GET /api/reports/* generates CSV exports (attendance, visits) for a manager-selected date range.
- GET /api/notifications surfaces manager alerts (a rep who didn't log in, an unresolved follow-up request) with clear/resolve actions.

5. Background Sweeps (APScheduler)

Sweep	Interval	Purpose
auto_cutoff_sweep	Every 15 min (30s after boot)	Force-closes a rep's day/visit left open past the configured cutoff hour
absence_check_sweep	Every 15 min (30s after boot)	Flags reps who never logged in for a business date, notifies managers (skips dates before the employee's account existed)
idempotency_cleanup	Every 1 hour	Purges expired Idempotency-Key records
employee_state_cache_sweep	Every 30 s	Expires the in-process employee-state cache

auto_cutoff_sweep and absence_check_sweep are guarded by a Postgres advisory lock (pg_try_advisory_lock) so that if the backend is ever scaled to more than one worker, only one instance actually runs each tick.

6. Offline-First Mobile Sync

- Every mutating request (check-in, check-out, dealer visit) is queued locally first and sent with a client-generated Idempotency-Key header.
- On network reconnect (monitored via @react-native-community/netinfo), the queue is replayed in order against the real backend.
- The backend's idempotency table deduplicates by key + employee + route, so a request that actually reached the server before the client lost connection is never double-applied when the queued copy is replayed.

Data Flow Summary

Rep

```
|
|--[Sign In]--> POST /api/auth/login --> JWT access + refresh tokens
|
|                               |
|                               expo-secure-store (mobile) / session (web)
|
|--[Start Day]--> GPS (expo-location) --> POST /api/attendance/checkin
|
|--[Navigate to Dealer]--> POST /api/navigation/compute --> Google Routes API
|
|                               |
|                               distance / duration / ETA / polyline saved
|
|--[Dealer Check-in]--> GPS --> POST /api/visits/login
|
|                               |
|                               haversine geofence check vs dealer radius
|
|--[Dealer Check-out]--> POST /api/visits/logout
|
|--[End Day]--> POST /api/attendance/checkout
|
(offline at any step -> queued locally with Idempotency-Key,
replayed automatically on reconnect)
```

Manager

```
|
|--[Dashboard]--> GET /api/dashboard/today --> live rep status, visit counts
|
|--[Map]--> GET /api/dashboard/map --> Google Maps JavaScript API (dealer/rep pins)
|
|--[Reports]--> GET /api/reports/* --> CSV export
|
'--[Notifications]--> GET /api/notifications --> absence alerts, follow-up requests
```

Deployment

Part	Platform	Trigger
Backend (backend-py)	Render (Blueprint via render.yaml)	Auto-deploy on push to main
Web (web/)	Vercel	Auto-deploy on push to main
Mobile (mobile/)	Expo EAS	eas update for JS-only changes (OTA); eas build for native/config changes